

American Option Pricing with Alternatives to Deep Neural Networks

Results Summary

September 2, 2026

This is an interim summary: 1 of 4 planned approximation methods is complete, and a cross-method comparison follows once more than one is finished.

The task is to predict the price of an American option from six inputs: asset price, maturity, rate, dividend, implied volatility, and the price of the corresponding European option. The dataset is proprietary and is not distributed with the code.

1 One-Dimensional Generalized Stochastic Sampling (gSS)

Following Antonov and Piterbarg (2021), gSS approximates the target as a linear combination of N product kernels centred on stochastically sampled nodes $z_n \in \mathbb{R}^D$ (their eqs. 25 and 30):

$$\tilde{f}(x) = \sum_{n=1}^N \beta_n \prod_{d=1}^D \psi((x_d - z_{n,d}) \omega_d), \quad (1)$$

$$\omega_d = \theta \kappa_d, \quad \kappa_d = \bar{g}_d / \Delta z, \quad (2)$$

$$\beta(\theta) = \arg \min_{\beta} \|\Psi(\theta)^\top \beta - Y\|_2^2 + \lambda \|\beta\|_2^2. \quad (3)$$

Here ψ is the inverse quadratic kernel, Δz the mean nearest-node distance, and $\bar{g}_d$ the relative directional risk magnitudes estimated from the gradients of a base fit. Only the coefficients β are regressed; the nodes are fixed and the single scalar θ is chosen by a bounded one-dimensional search. Ridge rather than plain least squares is required because the Gram matrix is near-degenerate at this node count.

The node count is then enriched from N to N' for evaluation, and θ re-calibrated on the dense set, because the optimum moves with node density: the value fitted on the sparse set is not the one to use at test time.

Fitted on 70 318 rows and evaluated on a held-out test set of 30 137 rows, 6 features.

On held-out data the enriched model reaches a relative L_2 error of 0.0032, against 0.0027 in sample. Both figures come from the same N' -node model, so the gap between them measures generalisation directly. It is small, and shows no sign of overfitting.

Interpretation. No baseline has been computed yet. The European option price is one of the six inputs and is itself a strong predictor of the American price, so some share of this accuracy comes from that input rather than from the method. Baselines follow with the cross-method comparison.

Simulation range	4
Nodes (fit model)	1 000
Nodes (test model)	5 000
Kernel	invquad
Node placement	random
Kernel scale θ	2
Ridge λ	1×10^{-6}
Floating-point precision	float64
Random seed	1

Test L_2 (dense)	0.0032
Train L_2 (dense)	0.0027
Test MAE (dense)	0.0017
Train MAE (dense)	0.0016
Train L_2 (fit model)	0.0053
Train MAE (fit model)	0.0031

Table 1: Configuration (left) and accuracy (right).

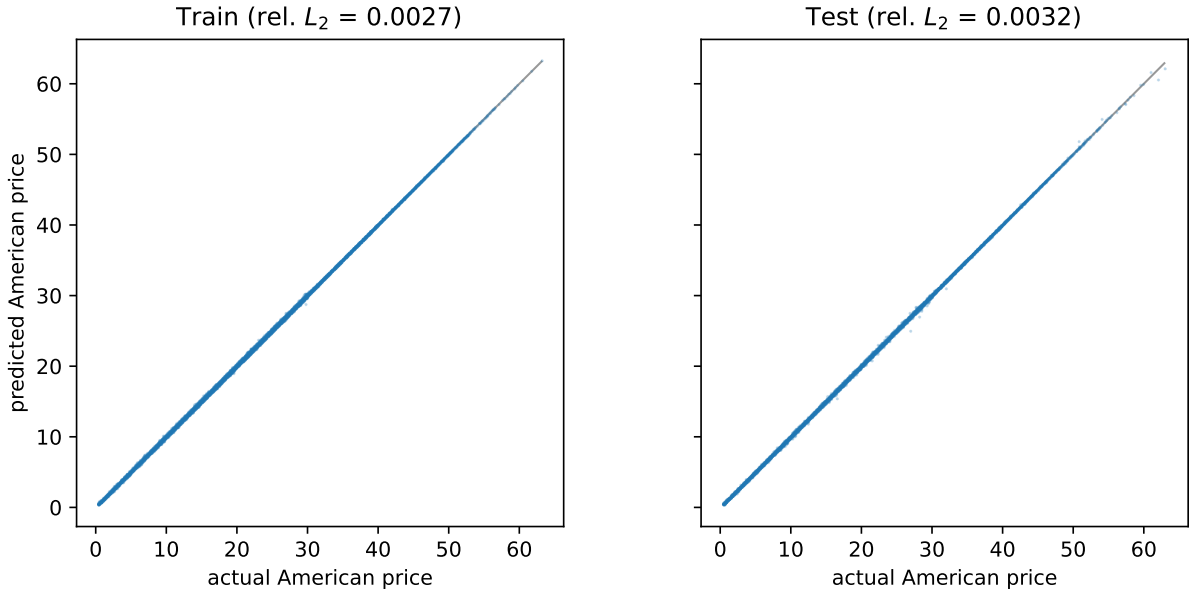


Figure 1: Predicted against actual American option price, with the $y = x$ reference in grey.

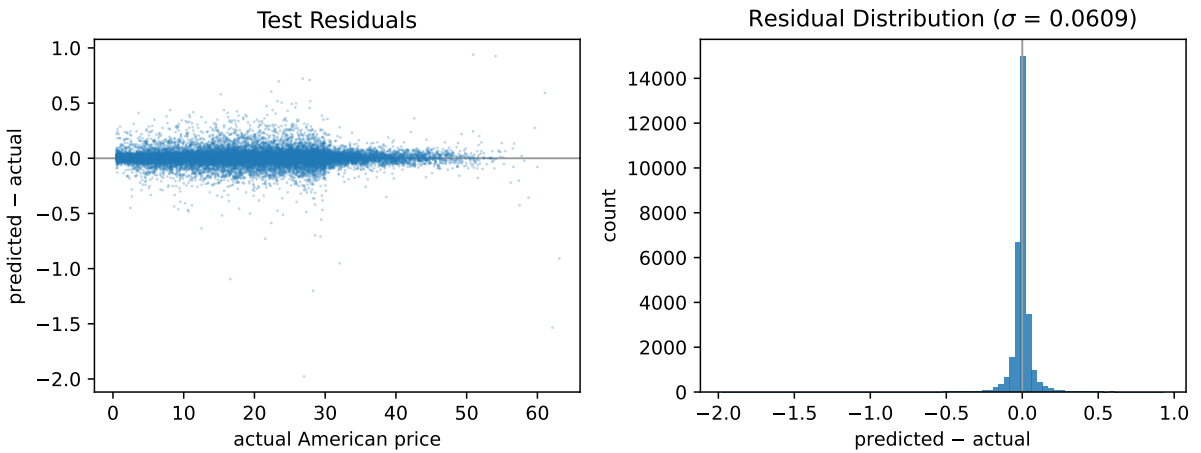


Figure 2: Test residuals against actual price, and their distribution.